# System Design Document

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EECS3311: Software Design

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# 1. Introduction

LionsList is a student-focused web-based marketplace designed exclusively for York University to buy and sell academic materials within the York University Community.

General marketplaces such as Kijiji and Facebook Marketplace do allow people to list textbooks and academic supplies but are not specifically aimed at university students. These platforms do not have the ability to filter by course, nor the verification of the users. Consequently, students frequently come across irrelevant listings, unreliable buyers, scams, and low value offers on these platforms.

The official York University bookstore is where students can get the academic materials they need. However, prices are usually fixed at high retail prices paired with a strict refund, return, and exchange policy. Online forums like the unofficial York University subreddit provide a medium for students to communicate, but lack structured listings, filter functionality, and user verification.

LionsList addresses these limitations by providing:

- York University Exclusive Access
- Course-code Filtering
- User Verification
- Searchable Academic Materials Listings
- Flexible Student Pricing

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## 2. System Environment

Runtime Environment

- Node.js (LTS version)
- MongoDB database
- Web browser for accessing the client application

Programming Technologies

- Frontend: React, Axios HTTP client, JavaScript / HTML / CSS
- Backend: Node.js, Express.js framework, Mongoose ODM for MongoDB
- Database: MongoDB

Project Structure

The repository contains two main components:

client/

- React frontend application
- Sends HTTP requests to backend API

server/

- Express backend API
- Handles business logic
- Communicates with MongoDB

Environment configuration is handled through a .env file in the server directory.

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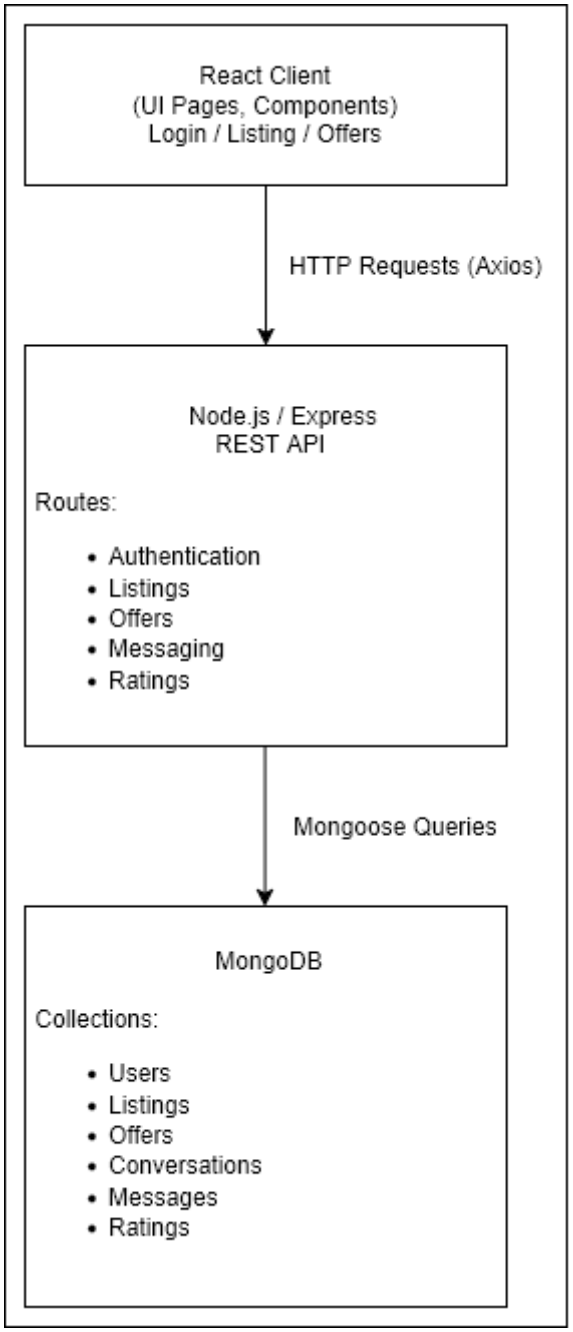
## 3. System Architecture




The LionsList system follows a three-tier web architecture consisting of a presentation layer, an application layer, and a data layer.

- The React frontend provides the user interface and communicates with the backend through HTTP requests using Axios.
- The Express backend processes these requests, applies business logic, and interacts with the MongoDB database through Mongoose models.
- MongoDB stores persistent application data such as users, listings, offers, conversations, and ratings.

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# 4. System Decomposition

## 4.1 Frontend Components

- Authentication pages (login / register)
- Browse listings page
- Listing detail page
- Create listing form
- User profile page
- Messaging interface
- Navigation bar

These components communicate with the backend using Axios API requests.

## 4.2 Backend Components

Backend functionality is implemented using Express routes and controllers.

Major backend modules include:

### Authentication

- Register users
- Login users
- Logout functionality

### Listings

- Create listings
- Edit listings
- Delete listings
- View listings

### Offers

- Submit offers
- Accept or reject offers
- Cancel offers

### Messaging

- Send messages
- Retrieve conversation threads

### Ratings

- Store ratings
- Calculate user reputation

### 4.3 Database Models

The system stores data in MongoDB using Mongoose schemas.

| Model | Description |
|---|---|
| User | Stores account credentials and rating statistics. |
| Listing | Represents an item listed for sale. |
| Offer | Represents an offer made by a buyer on a listing. |
| Conversation | Represents a message thread between buyer and seller. |
| Message | Individual messages inside a conversation. |
| Rating | Stores rating scores after completed transactions. |

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## 5. CRC Cards

| Class Name: User | |
|---|---|
| Parent Class (if any): None<br>Subclasses (if any): None | |
| Responsibilities:<br>● Register user accounts<br>● Authenticate login<br>● Store username and email<br>● Track user ratings<br>● View profile information | Collaborators:<br>● Listing<br>● Offer<br>● Conversation<br>● Rating |

| Class Name: Listing | |
|---|---|
| Parent Class (if any): None<br>Subclasses (if any): None | |
| Responsibilities:<br>● Store listing information<br>● Track listing available status<br>● Allow seller to edit or delete listing<br>● Display listing details to buyers | Collaborators:<br>● User<br>● Offer<br>● Conversation |

| Class Name: Offer | |
|---|---|
| Parent Class (if any): None<br>Subclasses (if any): None | |
| Responsibilities:<br>● Store offer amount<br>● Track offer status<br>● Allow buyers to cancel offers<br>● Allow sellers to accept or reject offers | Collaborators:<br>● User<br>● Listing<br>● Rating |

| Class Name: Conversation |
|---|

| Parent Class (if any): None<br>Subclasses (if any): None | |
|---|---|
| Responsibilities:<br>● Manage communication between buyer and seller<br>● Link conversation to listing<br>● Store participants | Collaborators:<br>● User<br>● Listing<br>● Message |

| Class Name: Message | |
|---|---|
| Parent Class (if any): None<br>Subclasses (if any): None | |
| Responsibilities:<br>● Store message text<br>● Track sender<br>● Track timestamp<br>● Attach message to conversation | Collaborators:<br>● Conversation<br>● User |

| Class Name: Rating | |
|---|---|
| Parent Class (if any): None<br>Subclasses (if any): None | |
| Responsibilities:<br>● Store rating score<br>● Link rating to completed transaction<br>● Update user reputation | Collaborators:<br>● User<br>● Offer |

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# 6. Error Handling Strategy

The LionsList system includes several mechanisms for detecting and responding to errors that may occur during normal operation.

These errors may originate from user input, authentication failures, network communication issues, or database problems. The system is designed so that failures are handled safely and communicated clearly to the user without exposing unnecessary internal system details.

Error handling occurs at both the backend (Express server) and frontend (React client) layers.

## 6.1 Invalid User Input

Users may enter invalid or incomplete information when registering accounts, creating listings, or submitting offers.

- Username already exists
- Email address not belonging to the York University domain
- Password shorter than the required length
- Missing listing information (title, description, price)

The backend implements input validation using Mongoose schema rules and server-side checks. For example, the User schema ensures:

- usernames are unique
- usernames only contain lowercase letters and numbers
- email addresses must end in *@yorku.ca* or *@my.yorku.ca*

When invalid input is detected:

1. The backend returns an HTTP 400 (Bad Request) response.
2. An error message is included in the response body.
3. The frontend displays the message to the user and prevents form submission until the issue is corrected.

This ensures that invalid data is not stored in the database.

## 6.2 Authentication and Authorization Errors

Certain actions require a user to be authenticated:

- creating a listing
- sending a message
- submitting an offer

If a user attempts to perform these actions without being logged in, the backend returns an HTTP 401 (Unauthorized) response.

The frontend detects this response and redirects the user to the login page.

Logout functionality removes the authentication token stored in the browser, ensuring the user can no longer access protected actions.

### 6.3 Database Errors

Database errors may occur if MongoDB becomes unavailable or if database operations fail.

- MongoDB connection failure
- duplicate username or email conflicts
- failed data writes

When a database error occurs:

1. The backend logs the error to the server console for debugging.
2. The API returns an HTTP 500 (Internal Server Error) response.
3. The frontend displays a generic error message to the user.

Sensitive system information is not exposed to the user for security reasons.

### 6.4 Network and API Communication Errors

Communication between the frontend and backend occurs through HTTP requests using Axios.

Possible failures include:

- server unreachable
- network timeouts
- incorrect API endpoint

When such errors occur:

1. Axios detects the request failure.
2. The frontend displays a message indicating that the request failed.
3. The user may retry the operation.

Example error messages:

- "Unable to connect to server."
- "Request failed. Please try again."

### 6.5 Data Integrity Protection

To maintain data integrity, the system enforces constraints at multiple levels.

- unique usernames and email addresses in the User schema
- listing ownership verification before editing or deleting a listing
- validation rules for message content and listing information

These safeguards ensure that invalid or unauthorized data cannot corrupt the system state.

### 6.6 Logging and Debugging

Server-side errors are logged in the backend console during development. These logs help developers identify issues such as failed database queries or unexpected exceptions.

In a production environment, these logs are able be redirected to a logging service or monitoring tool.

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